# Build and run

English version of ../../BUILD.md.

## Pinned versions

| Tool | Version | Source of truth |
|---|---|---|
| Python | 3.11+ (verified on 3.13) | render.yaml, README.md |
| Runtime dependencies | exact versions (==) | requirements.txt |
| Dev dependencies | exact versions (==) | requirements-dev.txt |
| httpx | 0.27.x | otherwise TestClient is incompatible with FastAPI 0.104 |

## Level A — source code

```
make doctor    # tools, dependencies, configuration, no secrets in code
make install
make lint
make test      # 15 tests; no network access
make docs-check
```

## Level B — run

```
cp .env.example .env     # set ADMIN_CODE
make run                 # uvicorn on port 8000
```

There is no build artefact: the application is interpreted.

## Level C — check a live instance

```
make smoke-test BASE_URL=http://localhost:8000
```

## Level D — CI

ci.yml runs three jobs: tests and lint (ruff + pytest), a runtime job that boots the application without optional configuration, and a docs job (make docs-check).

## Level E — documentation as artefacts

```
make docs-pdf    # Markdown -> PDF for both languages, output in docs/pdf/
```

The script `tools/docs_pdf.py` refuses to build if a document has no translation (`--check`), so the bilingual set cannot silently drift apart.